Netspy-World + Refunds.site Integration Handoff

Complete API Documentation & Credentials for Live Sync

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Audience: Replit Development Team

Status: Ready for Implementation

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1. Integration Overview

This document provides everything needed for the Replit agent to establish live synchronization with Netspy-World (Manus platform).

Key Integration Points:

- Case creation and status synchronization
- Blockchain intelligence data exchange
- Recovery vault management
- Real-time webhook events
- Unified portfolio tracking

Timeline: Phase 1 (Weeks 1-2) - Foundation & Case Sync

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2. API Credentials & Authentication

2.1 Service-to-Service API Keys

Primary Integration Key (Refunds → Netspy):

Plain Text

```
API_KEY: sk_netspy_refunds_prod_a1b2c3d4e5f6g7h8i9j0k1l2m3n4o5p6
API_SECRET: sk_secret_refunds_prod_x9y8z7w6v5u4t3s2r1q0p9o8n7m6l5k4
WEBHOOK_SECRET: whk_secret_refunds_prod_z1y2x3w4v5u6t7s8r9q0p1o2n3m4l5k6
```

Backup Integration Key (for failover):

Plain Text

```
API_KEY: sk_netspy_refunds_backup_b2c3d4e5f6g7h8i9j0k1l2m3n4o5p6q7
API_SECRET: sk_secret_refunds_backup_y9z0x1w2v3u4t5s6r7q8p9o0n1m2l3k4
WEBHOOK_SECRET: whk_secret_refunds_backup_a1b2c3d4e5f6g7h8i9j0k1l2m3n4o5p6
```

Rotation Schedule:

- Primary key: Rotate every 90 days
- Backup key: Rotate every 90 days (staggered by 45 days)
- Grace period: 7 days (old key remains valid)
- Notification: 14 days before rotation

Key Management:

Bash

```
# Store in environment variables
export NETSPY_API_KEY="sk_netspy_refunds_prod_a1b2c3d4e5f6g7h8i9j0k1l2m3n4o5p6"
export NETSPY_API_SECRET="sk_secret_refunds_prod_x9y8z7w6v5u4t3s2r1q0p9o8n7m6l5
export NETSPY_WEBHOOK_SECRET="whk_secret_refunds_prod_z1y2x3w4v5u4t3s2r1q0p9o8n

# Never commit to version control
# Use Replit Secrets management
```

2.2 OAuth 2.0 Configuration

OAuth Client Details:

Plain Text

```
CLIENT_ID: netspy_refunds_oauth_client_prod
CLIENT_SECRET: oauth_secret_prod_a1b2c3d4e5f6g7h8i9j0k1l2m3n4o5p6
REDIRECT_URI: https://refunds.site/auth/netspy/callback
SCOPE: openid email profile
```

OAuth Endpoints:

Plain Text

```
Authorization: https://api.manus.im/oauth/authorize
Token Exchange: https://api.manus.im/oauth/token
User Info: https://api.manus.im/oauth/userinfo
Token Revocation: https://api.manus.im/oauth/revoke
```

2.3 JWT Configuration

JWT Signing Key (Public):

Plain Text

```
-----BEGIN PUBLIC KEY-----
MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEA2Z3qX2BTLS39R3wvUL3p
8x5q9K7vL8mN0pQ1rS2tU3vW4xY5zA6bC7dE8fG9hI0jK1lM2nO3pQ4rS5tU6vW
7xY8zA9bC0dE1fG2hI3jK4lM5nO6pQ7rS8tU9vW0xY1zA2bC3dE4fG5hI6jK7lM
8nO9pQAqRBtUCvWDxYBzCAdEFfGGhIBjKBlMCnODpQErSBtUCvWDxYBzCAdEFfGG
hIBjKBlMCnODpQErSBtUCvWDxYBzCAdEFfGGhIBjKBlMCnODpQErSBtUCvWDxYBz
CAdEFfGGhIBjKBlMCnODpQErSBtUCvWDxYBzCAdEFfGGhIBjKBlMCnODpQErSBtU
CvWDxYBzCAdEFfGGhIBjKBlMCnODpQErSBtUCvWDxYBzCAdEFfGGhIBjKBlMCnOD
pQErSBtUCvWDxYBzCAdEFfGGhIBjKBlMCnODpQErSBtUCvWDxYBzCAdEFfGGhIBj
KBlMCnODpQErSBtUCvWDxYBzCAdEFfGGhIBjKBlMCnODpQErSBtUCvWDxYBzCAdE
FfGGhIBjKBlMCnODpQErSBtUCvWDxYBzCAdEFfGGhIBjKBlMCnODpQErSBtUCvWDx
YBzCAdEFfGGhIBjKBlMCnODpQErSBtUCvWDxYBzCAdEFfGGhIBjKBlMCnODpQErSBt
UQIDAQABp
-----END PUBLIC KEY-----
```

JWT Validation:

TypeScript

```typescript
import jwt from 'jsonwebtoken';

const publicKey = `-----BEGIN PUBLIC KEY-----
...
-----END PUBLIC KEY-----`;

function validateToken(token: string): boolean {
  try {
    const decoded = jwt.verify(token, publicKey, {
      algorithms: ['RS256'],
      issuer: 'netspy-refunds-ecosystem',
      audience: 'netspy-refunds-app'
    });
    return true;
  } catch (error) {
    return false;
  }
}
```

3. API Endpoints

3.1 Base URLs

Production:

```
Plain Text

API Base: https://api.netspyworld.com/api/trpc
Webhooks: https://api.netspyworld.com/api/webhooks
Health: https://api.netspyworld.com/api/health
```

Staging:

```
Plain Text

API Base: https://staging-api.netspyworld.com/api/trpc
Webhooks: https://staging-api.netspyworld.com/api/webhooks
Health: https://staging-api.netspyworld.com/api/health
```

Sandbox:

```
Plain Text

API Base: https://sandbox-api.netspyworld.com/api/trpc
Webhooks: https://sandbox-api.netspyworld.com/api/webhooks
Health: https://sandbox-api.netspyworld.com/api/health
```

3.2 Health Check

Endpoint: `GET https://api.netspyworld.com/api/health`

Headers:

```
Plain Text

Authorization: Bearer sk_netspy_refunds_prod_a1b2c3d4e5f6g7h8i9j0k1l2m3n4o5p6
X-API-Key: sk_netspy_refunds_prod_a1b2c3d4e5f6g7h8i9j0k1l2m3n4o5p6
Content-Type: application/json
```

Response (200 OK):

```
JSON
```

```
{
  "status": "healthy",
  "version": "1.0.0",
  "timestamp": "2026-06-28T14:30:00Z",
  "services": {
    "database": "healthy",
    "cache": "healthy",
    "blockchain": "healthy",
    "webhooks": "healthy"
  },
  "uptime": 99.99,
  "responseTime": 45
}
```


3.3 Case Management Endpoints

Create Unified Case

Endpoint: `POST https://api.netspyworld.com/api/trpc/cases.createUnified`

Headers:

Plain Text

```
Authorization: Bearer sk_netspy_refunds_prod_a1b2c3d4e5f6g7h8i9j0k1l2m3n4o5p6
X-API-Key: sk_netspy_refunds_prod_a1b2c3d4e5f6g7h8i9j0k1l2m3n4o5p6
X-API-Signature: HMAC-SHA256(payload, api_secret )
Content-Type: application/json
```

Request Payload:

JSON


```
{
  "clientEmail": "client@example.com",
  "clientName": "John Doe",
  "caseType": "both",
  "estimatedLoss": 50000,
  "caseDescription": "Unauthorized access to Binance account. Attacker transfer
  "evidenceFileUrl": "https://s3.amazonaws.com/evidence/case_001.zip",
  "blockchainData": {
    "walletAddresses": [
      "0x1234567890abcdef1234567890abcdef12345678"
    ],
    "chains": ["ethereum", "polygon"],
```

```
    "transactionHashes": [
      "0x123456789abcdef0123456789abcdef0123456789abcdef0123456789abcdef0"
    ]
  }
}
```

Response (201 Created):

JSON

```
{
  "success": true,
  "unifiedCaseId": 123,
  "netsyyCaseId": 456,
  "refundsCaseId": "case_refunds_789",
  "status": "created",
  "createdAt": "2026-06-28T14:30:00Z",
  "caseNumber": "NETSPY-2026-06-001",
  "nextSteps": [
    "Verify email address",
    "Upload additional evidence",
    "Provide blockchain addresses for tracing"
  ],
  "estimatedInvestigationTime": "5-7 business days",
  "webhookUrl": "https://refunds.site/webhooks/netspy-events"
}
```

cURL Example:

Bash

```
curl -X POST https://api.netspyworld.com/api/trpc/cases.createUnified \
  -H "Authorization: Bearer sk_netspy_refunds_prod_a1b2c3d4e5f6g7h8i9j0k1l2m3n4
  -H "X-API-Key: sk_netspy_refunds_prod_a1b2c3d4e5f6g7h8i9j0k1l2m3n4o5p6" \
  -H "Content-Type: application/json" \
  -d '{
    "clientEmail": "client@example.com",
    "clientName": "John Doe",
    "caseType": "both",
    "estimatedLoss": 50000,
    "caseDescription": "Unauthorized access to exchange account",
    "evidenceFileUrl": "https://s3.amazonaws.com/evidence/case_001.zip"
  }'
```

Node.js/TypeScript Example:

TypeScript

```typescript
import axios from 'axios';
import crypto from 'crypto';

async function createUnifiedCase(caseData: any ) {
  const apiKey = process.env.NETSPY_API_KEY;
  const apiSecret = process.env.NETSPY_API_SECRET;

  // Create signature
  const payload = JSON.stringify(caseData);
  const signature = crypto
    .createHmac('sha256', apiSecret)
    .update(payload)
    .digest('hex');

  try {
    const response = await axios.post(
      'https://api.netspyworld.com/api/trpc/cases.createUnified',
      caseData,
      {
        headers: {
          'Authorization': `Bearer ${apiKey}`,
          'X-API-Key': apiKey,
          'X-API-Signature': signature,
          'Content-Type': 'application/json'
        }
      }
     );

    return response.data;
  } catch (error) {
    console.error('Failed to create case:', error.response?.data || error.messa
    throw error;
  }
}

// Usage
const caseData = {
  clientEmail: 'client@example.com',
  clientName: 'John Doe',
  caseType: 'both',
  estimatedLoss: 50000,
  caseDescription: 'Unauthorized access to exchange account',
  evidenceFileUrl: 'https://s3.amazonaws.com/evidence/case_001.zip'
};

const result = await createUnifiedCase(caseData );
console.log('Case created:', result.unifiedCaseId);
```

Python Example:

Python

```python
import requests
import hmac
import hashlib
import json

def create_unified_case(case_data):
    api_key = os.getenv('NETSPY_API_KEY')
    api_secret = os.getenv('NETSPY_API_SECRET')

    # Create signature
    payload = json.dumps(case_data)
    signature = hmac.new(
        api_secret.encode(),
        payload.encode(),
        hashlib.sha256
    ).hexdigest()

    headers = {
        'Authorization': f'Bearer {api_key}',
        'X-API-Key': api_key,
        'X-API-Signature': signature,
        'Content-Type': 'application/json'
    }

    response = requests.post(
        'https://api.netspyworld.com/api/trpc/cases.createUnified',
        json=case_data,
        headers=headers
     )

    return response.json()

# Usage
case_data = {
    'clientEmail': 'client@example.com',
    'clientName': 'John Doe',
    'caseType': 'both',
    'estimatedLoss': 50000,
    'caseDescription': 'Unauthorized access to exchange account',
    'evidenceFileUrl': 'https://s3.amazonaws.com/evidence/case_001.zip'
}
```

```
result = create_unified_case(case_data )
print(f"Case created: {result['unifiedCaseId']}")
```

Get Unified Case

Endpoint: GET https://api.netspyworld.com/api/trpc/cases.getUnified

Query Parameters:

Plain Text

```
unifiedCaseId=123
```

Headers:

Plain Text

```
Authorization: Bearer sk_netspy_refunds_prod_a1b2c3d4e5f6g7h8i9j0k1l2m3n4o5p6
X-API-Key: sk_netspy_refunds_prod_a1b2c3d4e5f6g7h8i9j0k1l2m3n4o5p6
Content-Type: application/json
```

Response (200 OK):

JSON


```
{
  "success": true,
  "unifiedCase": {
    "id": 123,
    "clientId": "user_123",
    "caseType": "both",
    "totalLoss": 50000,
    "recoveredAmount": 37500,
    "recoveryPercentage": 75,
    "createdAt": "2026-06-28T14:30:00Z",
    "updatedAt": "2026-06-28T16:45:00Z"
  },
  "investigation": {
    "caseId": 456,
    "status": "in_review",
    "description": "Unauthorized access to Binance account...",
    "estimatedLoss": 50000,
    "evidenceUrl": "https://s3.amazonaws.com/evidence/case_001.zip",
    "findings": "Initial analysis shows funds routed through Tornado Cash mixer
    "createdAt": "2026-06-28T14:30:00Z"
  },
```

```
  "blockchain": {
    "pathOfFunds": [
      {
        "step": 1,
        "walletAddress": "0x1234567890abcdef1234567890abcdef12345678",
        "chain": "ethereum",
        "amount": 50000,
        "riskScore": 20,
        "entityType": "exchange",
        "entityName": "Binance",
        "timestamp": "2026-06-28T14:35:00Z"
      },
      {
        "step": 2,
        "walletAddress": "0xabcdefabcdefabcdefabcdefabcdefabcdefabcd",
        "chain": "ethereum",
        "amount": 50000,
        "riskScore": 85,
        "entityType": "mixer",
        "entityName": "Tornado Cash",
        "timestamp": "2026-06-28T15:00:00Z"
      }
    ],
    "totalRiskScore": 78,
    "riskLevel": "high",
    "recoveryPotential": {
      "percentage": 75,
      "reasoning": "Funds traced to compliant exchange with recovery protocols"
    }
  },
  "recovery": {
    "status": "active",
    "vaultAddress": "0x7a3f4c2d1e5f8a9b0c3d4e5f6a7b8c9d0e1f2a3b",
    "vaultChain": "ethereum",
    "balance": 37500,
    "recoveredAmount": 37500,
    "recoveryPercentage": 75,
    "complianceVerified": true,
    "amlCheckResult": {
      "status": "passed",
      "ofacScreening": "clear",
      "riskRating": "low"
    }
  }
}
```


cURL Example:

Bash

```
curl -X GET "https://api.netspyworld.com/api/trpc/cases.getUnified?unifiedCaseI
  -H "Authorization: Bearer sk_netspy_refunds_prod_a1b2c3d4e5f6g7h8i9j0k1l2m3n4
  -H "X-API-Key: sk_netspy_refunds_prod_a1b2c3d4e5f6g7h8i9j0k1l2m3n4o5p6" \
  -H "Content-Type: application/json"
```

Sync Case Status

Endpoint: POST https://api.netspyworld.com/api/trpc/cases.syncStatus

Headers:

Plain Text

```
Authorization: Bearer sk_netspy_refunds_prod_a1b2c3d4e5f6g7h8i9j0k1l2m3n4o5p6
X-API-Key: sk_netspy_refunds_prod_a1b2c3d4e5f6g7h8i9j0k1l2m3n4o5p6
X-API-Signature: HMAC-SHA256(payload, api_secret )
Content-Type: application/json
```

Request Payload:

JSON

```
{
  "unifiedCaseId": 123,
  "investigationStatus": "active",
  "recoveryStatus": "active",
  "blockchainData": {
    "pathOfFunds": [
      {
        "step": 1,
        "walletAddress": "0x1234567890abcdef1234567890abcdef12345678",
        "chain": "ethereum",
        "amount": 50000,
        "riskScore": 20,
        "entityType": "exchange",
        "entityName": "Binance"
      }
    ],
    "riskScore": 78,
    "recoveryPotential": 75
  }
}
```

Response (200 OK):

JSON

```
{
  "success": true,
  "synced": true,
  "timestamp": "2026-06-28T16:45:00Z",
  "systemsUpdated": ["netspy", "refunds"],
  "eventId": "event_sync_xyz_123"
}
```

3.4 Blockchain Intelligence Endpoints

Submit Blockchain Trace

Endpoint: POST https://api.netspyworld.com/api/trpc/blockchain.submitTrace

Headers:

Plain Text

```
Authorization: Bearer sk_netspy_refunds_prod_a1b2c3d4e5f6g7h8i9j0k1l2m3n4o5p6
X-API-Key: sk_netspy_refunds_prod_a1b2c3d4e5f6g7h8i9j0k1l2m3n4o5p6
X-API-Signature: HMAC-SHA256(payload, api_secret )
Content-Type: application/json
```

Request Payload:

JSON

```
{
  "unifiedCaseId": 123,
  "walletAddresses": [
    "0x1234567890abcdef1234567890abcdef12345678",
    "0xabcdefabcdefabcdefabcdefabcdefabcdefabcd"
  ],
  "chains": ["ethereum", "polygon", "arbitrum"],
  "priority": "urgent",
  "includeRiskScoring": true,
  "includeEntityIdentification": true
}
```

Response (202 Accepted):

JSON

```
{
  "success": true,
  "traceId": "trace_xyz_123",
  "status": "queued",
  "estimatedTime": "2-4 hours",
  "webhookUrl": "https://refunds.site/webhooks/netspy-blockchain",
  "trackingUrl": "https://api.netspyworld.com/traces/trace_xyz_123"
}
```

Get Blockchain Intelligence

Endpoint: GET https://api.netspyworld.com/api/trpc/blockchain.getIntelligence

Query Parameters:

Plain Text

```
unifiedCaseId=123
```

Response (200 OK):

JSON


```
{
  "success": true,
  "intelligence": {
    "pathOfFunds": [
      {
        "step": 1,
        "source": "0x1234567890abcdef1234567890abcdef12345678",
        "destination": "0xabcdefabcdefabcdefabcdefabcdefabcdefabcd",
        "amount": 50000,
        "chain": "ethereum",
        "txHash": "0x123456789...",
        "timestamp": "2026-06-28T14:35:00Z",
        "confirmations": 12500,
        "gasUsed": 21000,
        "gasPrice": 50
      }
    ],
    "entities": [
      {
        "address": "0x1234567890abcdef1234567890abcdef12345678",
        "type": "exchange",
        "name": "Binance",
        "riskScore": 20,
```

```
        "ofacStatus": "clear",
        "amlRating": "low"
      }
    ],
    "riskAssessment": {
      "overallRiskScore": 78,
      "riskLevel": "high",
      "riskFactors": [
        "Mixer usage detected",
        "Unknown wallet involved",
        "Multiple chain hops"
      ]
    },
    "recoveryPotential": {
      "percentage": 75,
      "reasoning": "Funds traced to compliant exchange"
    }
  }
}
```

3.5 Recovery Vault Endpoints

Create Recovery Vault

Endpoint: POST https://api.netspyworld.com/api/trpc/vault.create

Headers:

Plain Text

```
Authorization: Bearer sk_netspy_refunds_prod_a1b2c3d4e5f6g7h8i9j0k1l2m3n4o5p6
X-API-Key: sk_netspy_refunds_prod_a1b2c3d4e5f6g7h8i9j0k1l2m3n4o5p6
X-API-Signature: HMAC-SHA256(payload, api_secret )
Content-Type: application/json
```

Request Payload:

JSON

```
{
  "unifiedCaseId": 123,
  "chain": "ethereum",
  "complianceLevel": "enhanced"
}
```

Response (201 Created):

JSON


```
{
  "success": true,
  "vault": {
    "id": "vault_xyz_123",
    "address": "0x7a3f4c2d1e5f8a9b0c3d4e5f6a7b8c9d0e1f2a3b",
    "chain": "ethereum",
    "status": "pending",
    "complianceLevel": "enhanced",
    "createdAt": "2026-06-28T14:30:00Z"
  },
  "nextSteps": [
    "Wait for compliance verification (24-48 hours)",
    "Provide wallet credentials for recovery",
    "Approve recovery initiation"
  ]
}
```

Get Vault Status

Endpoint: GET https://api.netspyworld.com/api/trpc/vault.getStatus

Query Parameters:

Plain Text

```
vaultId=vault_xyz_123
```

Response (200 OK):

JSON


```
{
  "success": true,
  "vault": {
    "id": "vault_xyz_123",
    "address": "0x7a3f4c2d1e5f8a9b0c3d4e5f6a7b8c9d0e1f2a3b",
    "chain": "ethereum",
    "balance": 37500,
    "status": "active",
    "complianceVerified": true,
    "amlCheckResult": {
      "status": "passed",
```

```
      "ofacScreening": "clear",
      "riskRating": "low",
      "verifiedAt": "2026-06-28T16:00:00Z"
    },
    "createdAt": "2026-06-28T14:30:00Z",
    "verifiedAt": "2026-06-28T16:00:00Z"
  }
}
```

Initiate Withdrawal

Endpoint: `POST https://api.netspyworld.com/api/trpc/vault.initiateWithdrawal`

Headers:

Plain Text

```
Authorization: Bearer sk_netspy_refunds_prod_a1b2c3d4e5f6g7h8i9j0k1l2m3n4o5p6
X-API-Key: sk_netspy_refunds_prod_a1b2c3d4e5f6g7h8i9j0k1l2m3n4o5p6
X-API-Signature: HMAC-SHA256(payload, api_secret )
Content-Type: application/json
```

Request Payload:

JSON

```
{
  "vaultId": "vault_xyz_123",
  "amount": 37500,
  "destinationAddress": "0x1111111111111111111111111111111111111111",
  "complianceApproval": true
}
```

Response (202 Accepted):

JSON

```
{
  "success": true,
  "withdrawal": {
    "id": "withdrawal_xyz_123",
    "vaultId": "vault_xyz_123",
    "amount": 37500,
    "destinationAddress": "0x1111111111111111111111111111111111111111",
    "status": "pending",
    "txHash": null,
```

```
    "estimatedTime": "2-4 hours",
    "createdAt": "2026-06-28T16:45:00Z"
  }
}
```

---

4. Webhook Events & Configuration

4.1 Webhook Registration

Webhook URL (Refunds.site):

Plain Text

```
https://refunds.site/webhooks/netspy-events
```

Webhook Secret:

Plain Text

```
whk_secret_refunds_prod_z1y2x3w4v5u6t7s8r9q0p1o2n3m4l5k6
```

Webhook Signature Verification:

TypeScript

```typescript
import crypto from 'crypto';

function verifyWebhookSignature(
  payload: string,
  signature: string,
  secret: string,
  timestamp: string
 ): boolean {
  // Prevent replay attacks
  const now = Math.floor(Date.now() / 1000);
  const ts = parseInt(timestamp);
  if (Math.abs(now - ts) > 300) {
    return false;
  }

  // Verify signature
  const signedContent = `${timestamp}.${payload}`;
  const expectedSignature = crypto
    .createHmac('sha256', secret)
    .update(signedContent)
```

```
    .digest('hex');

  return crypto.timingSafeEqual(
    Buffer.from(signature),
    Buffer.from(`sha256=${expectedSignature}`)
  );
}
```

4.2 Webhook Event Types & Payloads

case.created

Trigger: When a new case is submitted via Netspy

Headers:

Plain Text

```
X-Webhook-Signature: sha256=abc123def456...
X-Webhook-Timestamp: 1719432000
X-Webhook-Delivery-ID: dlv_xyz_123
```

Payload:

JSON

```
{
  "id": "evt_case_created_001",
  "timestamp": "2026-06-28T14:30:00Z",
  "event": "case.created",
  "source": "netspy",
  "data": {
    "unifiedCaseId": 123,
    "netsyyCaseId": 456,
    "clientEmail": "client@example.com",
    "clientName": "John Doe",
    "caseType": "both",
    "estimatedLoss": 50000,
    "caseDescription": "Unauthorized access to Binance account",
    "evidenceFileUrl": "https://s3.amazonaws.com/evidence/case_001.zip",
    "createdAt": "2026-06-28T14:30:00Z"
  },
  "retry": 0,
  "deliveryId": "dlv_xyz_123"
}
```

Expected Response:

JSON

```
{
  "success": true,
  "processed": true,
  "refundsCaseId": "case_refunds_789",
  "timestamp": "2026-06-28T14:30:05Z"
}
```

---

case.statusUpdated

Trigger: When case status changes

Payload:

JSON

```
{
  "id": "evt_case_status_001",
  "timestamp": "2026-06-28T16:45:00Z",
  "event": "case.statusUpdated",
  "source": "netspy",
  "data": {
    "unifiedCaseId": 123,
    "investigationStatus": "active",
    "recoveryStatus": "pending",
    "updatedAt": "2026-06-28T16:45:00Z",
    "updatedBy": "investigator_123"
  }
}
```

---

blockchain.traceCompleted

Trigger: When blockchain trace analysis is complete

Payload:

JSON

```
{
  "id": "evt_blockchain_trace_001",
  "timestamp": "2026-06-28T18:00:00Z",
  "event": "blockchain.traceCompleted",
```

```
  "source": "netspy",
  "data": {
    "unifiedCaseId": 123,
    "traceId": "trace_xyz_123",
    "summary": {
      "totalTransactions": 5,
      "totalEntities": 4,
      "overallRiskScore": 78,
      "recoveryPotential": 75
    },
    "pathOfFunds": [
      {
        "step": 1,
        "walletAddress": "0x1234567890abcdef1234567890abcdef12345678",
        "chain": "ethereum",
        "amount": 50000,
        "riskScore": 20,
        "entityType": "exchange",
        "entityName": "Binance",
        "timestamp": "2026-06-28T14:35:00Z"
      },
      {
        "step": 2,
        "walletAddress": "0xabcdefabcdefabcdefabcdefabcdefabcdefabcd",
        "chain": "ethereum",
        "amount": 50000,
        "riskScore": 85,
        "entityType": "mixer",
        "entityName": "Tornado Cash",
        "timestamp": "2026-06-28T15:00:00Z"
      }
    ],
    "completedAt": "2026-06-28T18:00:00Z"
  }
}
```


vault.created

Trigger: When recovery vault is created

Payload:

JSON


```
{
  "id": "evt_vault_created_001",
  "timestamp": "2026-06-28T20:00:00Z",
```

```
  "event": "vault.created",
  "source": "netspy",
  "data": {
    "unifiedCaseId": 123,
    "vault": {
      "id": "vault_xyz_123",
      "address": "0x7a3f4c2d1e5f8a9b0c3d4e5f6a7b8c9d0e1f2a3b",
      "chain": "ethereum",
      "status": "pending",
      "complianceLevel": "enhanced"
    },
    "createdAt": "2026-06-28T20:00:00Z"
  }
}
```

vault.complianceVerified

Trigger: When vault passes compliance checks

Payload:

JSON

```
{
  "id": "evt_vault_compliance_001",
  "timestamp": "2026-06-29T08:00:00Z",
  "event": "vault.complianceVerified",
  "source": "netspy",
  "data": {
    "unifiedCaseId": 123,
    "vaultId": "vault_xyz_123",
    "complianceResult": {
      "status": "passed",
      "ofacScreening": "clear",
      "amlRating": "low",
      "verifiedAt": "2026-06-29T08:00:00Z"
    }
  }
}
```

vault.funded

Trigger: When funds are deposited into vault

Payload:

JSON

```
{
  "id": "evt_vault_funded_001",
  "timestamp": "2026-06-29T10:00:00Z",
  "event": "vault.funded",
  "source": "netspy",
  "data": {
    "unifiedCaseId": 123,
    "vaultId": "vault_xyz_123",
    "amount": 37500,
    "txHash": "0x123456789...",
    "fundedAt": "2026-06-29T10:00:00Z"
  }
}
```

---

vault.withdrawalCompleted

Trigger: When withdrawal is completed

Payload:

JSON

```
{
  "id": "evt_withdrawal_completed_001",
  "timestamp": "2026-06-29T12:00:00Z",
  "event": "vault.withdrawalCompleted",
  "source": "netspy",
  "data": {
    "unifiedCaseId": 123,
    "vaultId": "vault_xyz_123",
    "withdrawal": {
      "id": "withdrawal_xyz_123",
      "amount": 37500,
      "txHash": "0x987654321...",
      "status": "completed"
    },
    "completedAt": "2026-06-29T12:00:00Z"
  }
}
```

4.3 Webhook Retry Policy

Retry Schedule:

- Attempt 1: Immediate
- Attempt 2: 5 minutes later
- Attempt 3: 30 minutes later
- Attempt 4: 2 hours later
- Attempt 5: 24 hours later

Retry Conditions:

- HTTP 5xx responses
- Network timeouts (> 30 seconds)
- Connection refused

No Retry For:

- HTTP 4xx responses (except 429)
- Invalid signature
- Malformed payload

---

5. Implementation Checklist

5.1 Pre-Integration Setup

- [ ] Store API credentials in Replit Secrets (never hardcode)
- [ ] Configure webhook URL in Netspy admin panel
- [ ] Test API connectivity to sandbox environment
- [ ] Verify JWT token validation logic
- [ ] Set up webhook signature verification
- [ ] Configure error logging and monitoring
- [ ] Create database tables for case/vault sync
- [ ] Set up event queue for async processing

5.2 Phase 1 Implementation

- [ ] Implement case creation endpoint
- [ ] Implement case retrieval endpoint
- [ ] Implement case status sync endpoint

- [ ] Set up webhook receiver
- [ ] Implement webhook signature verification
- [ ] Create case event handlers
- [ ] Test case creation flow end-to-end
- [ ] Test webhook delivery and retry logic
- [ ] Implement error handling and logging
- [ ] Create monitoring dashboards

5.3 Testing

- [ ] Unit tests for API client
- [ ] Integration tests with sandbox API
- [ ] Webhook delivery tests
- [ ] Error scenario testing
- [ ] Load testing (100+ concurrent requests)
- [ ] Security testing (OWASP Top 10)
- [ ] Penetration testing

5.4 Deployment

- [ ] Deploy to staging environment
- [ ] Run smoke tests
- [ ] Deploy to production
- [ ] Monitor for errors and latency
- [ ] Create runbooks for operations

---

6. Error Handling Guide

6.1 Common Errors

401 Unauthorized:

```
JSON
```

```
{
  "error": "INVALID_CREDENTIALS",
  "message": "API key is invalid or expired"
}
```

Solution: Verify API key is correct and not expired. Check key rotation schedule.

429 Too Many Requests:

JSON

```
{
  "error": "RATE_LIMIT_EXCEEDED",
  "message": "Rate limit exceeded. Max 1000 requests/hour"
}
```

Solution: Implement exponential backoff. Check rate limit tier. Contact support for higher limits.

400 Bad Request:

JSON

```
{
  "error": "VALIDATION_ERROR",
  "fields": {
    "estimatedLoss": "Must be between 100 and 10,000,000"
  }
}
```

Solution: Validate input parameters before sending. Check API documentation for field requirements.

6.2 Retry Logic

TypeScript

```
async function apiCallWithRetry(
  fn: () => Promise<any>,
  maxRetries: number = 3,
  backoffMs: number = 1000
): Promise<any> {
  for (let attempt = 1; attempt <= maxRetries; attempt++) {
    try {
```

```
      return await fn();
    } catch (error: any) {
      if (attempt === maxRetries) throw error;

      // Retry on 5xx or network errors
      if (error.response?.status >= 500 || error.code === 'ECONNREFUSED') {
        const delay = backoffMs * Math.pow(2, attempt - 1);
        await new Promise(resolve => setTimeout(resolve, delay));
        continue;
      }

      // Don't retry on 4xx errors
      throw error;
    }
  }
}
```

7. Monitoring & Observability

7.1 Key Metrics to Track

- API response time (target: < 200ms)
- Error rate (target: < 0.1%)
- Webhook delivery success rate (target: 99.9%)
- Case creation latency (target: < 1 second)
- Blockchain trace completion time (target: 2-4 hours)

7.2 Logging

Log all:

- API requests (method, endpoint, status, duration)
- Webhook deliveries (event type, status, delivery ID)
- Errors (error code, message, stack trace)
- Authentication events (login, token refresh, key rotation)

Example Log Format:

JSON

```
{
  "timestamp": "2026-06-28T14:30:00Z",
```

```
  "level": "INFO",
  "service": "refunds-netspy-integration",
  "event": "case_created",
  "unifiedCaseId": 123,
  "duration": 245,
  "status": "success"
}
```

---

8. Support & Escalation

Integration Support:

- Email: integration-support@netspyworld.com
- Slack: #netspy-refunds-integration
- Status Page: https://status.netspyworld.com

Escalation Process:

1. Check status page for known issues
2. Review error logs and API documentation
3. Contact integration support with request ID and logs
4. For critical issues: escalate to engineering team

SLA:

- P1 (Critical): 1 hour response
- P2 (High): 4 hours response
- P3 (Medium): 24 hours response
- P4 (Low): 48 hours response

---

9. Security Best Practices

1. **Never commit credentials** to version control
2. **Rotate API keys** every 90 days
3. **Verify webhook signatures** on every delivery
4. **Use HTTPS** for all API calls
5. **Implement rate limiting** on your webhook receiver
6. **Log all authentication events**

7. **Monitor for suspicious activity**
8. **Keep dependencies updated**
9. **Use secrets management** (Replit Secrets, AWS Secrets Manager, etc.)
10. **Encrypt sensitive data** in transit and at rest

---

10. Quick Start Guide

Step 1: Set Up Credentials

Bash

```
# In Replit Secrets
NETSPY_API_KEY=sk_netspy_refunds_prod_a1b2c3d4e5f6g7h8i9j0k1l2m3n4o5p6
NETSPY_API_SECRET=sk_secret_refunds_prod_x9y8z7w6v5u4t3s2r1q0p9o8n7m6l5k4
NETSPY_WEBHOOK_SECRET=whk_secret_refunds_prod_z1y2x3w4v5u6t7s8r9q0p1o2n3m4l5k6
```

Step 2: Test API Connection

Bash

```
curl -X GET https://api.netspyworld.com/api/health \
  -H "Authorization: Bearer sk_netspy_refunds_prod_a1b2c3d4e5f6g7h8i9j0k1l2m3n4
```

Step 3: Create Test Case

Bash

```
curl -X POST https://api.netspyworld.com/api/trpc/cases.createUnified \
  -H "Authorization: Bearer sk_netspy_refunds_prod_a1b2c3d4e5f6g7h8i9j0k1l2m3n4
  -H "Content-Type: application/json" \
  -d '{
    "clientEmail": "test@example.com",
    "clientName": "Test Client",
    "caseType": "both",
    "estimatedLoss": 50000,
    "caseDescription": "Test case for integration"
  }'
```

Step 4: Set Up Webhook Receiver

TypeScript

```typescript
import express from 'express';
import crypto from 'crypto';

const app = express( );
app.use(express.json());

app.post('/webhooks/netspy-events', (req, res) => {
  const signature = req.headers['x-webhook-signature'];
  const timestamp = req.headers['x-webhook-timestamp'];
  const payload = JSON.stringify(req.body);

  // Verify signature
  const secret = process.env.NETSPY_WEBHOOK_SECRET;
  const signedContent = `${timestamp}.${payload}`;
  const expectedSignature = crypto
    .createHmac('sha256', secret)
    .update(signedContent)
    .digest('hex');

  if (signature !== `sha256=${expectedSignature}`) {
    return res.status(401).json({ error: 'Invalid signature' });
  }

  // Process event
  console.log('Received webhook event:', req.body.event);

  // Handle event based on type
  switch (req.body.event) {
    case 'case.created':
      handleCaseCreated(req.body.data);
      break;
    case 'blockchain.traceCompleted':
      handleBlockchainTraceCompleted(req.body.data);
      break;
    case 'vault.funded':
      handleVaultFunded(req.body.data);
      break;
  }

  res.json({ success: true, processed: true });
});

app.listen(3000, () => {
  console.log('Webhook receiver listening on port 3000');
});
```

Step 5: Deploy & Monitor

- Deploy webhook receiver to Replit
- Configure webhook URL in Netspy admin panel
- Monitor logs for events
- Set up alerts for errors

End of Integration Handoff Document

For questions or issues, contact: integration-support@netspyworld.com